11.1 Circumference and Arc Length

Name _____ Period ____ Date _____

Geometry 11.1 (30-minute plan) · Student sheet

I can…

- find the circumference of a circle and use it to solve problems.
- find the length of an arc and use arc lengths to find measures.
- convert between degree and radian measure.

Vocabulary · fill in the missing word

Circumference — the _____ around a circle

Arc length — a portion of the _____ of a circle

Radian — the _____ of a central angle whose arc is the same length as the radius

Part 1 · Notes and examples (fill in each note, then work the example)

Circumference: C = _____ = 2πr

Example 1

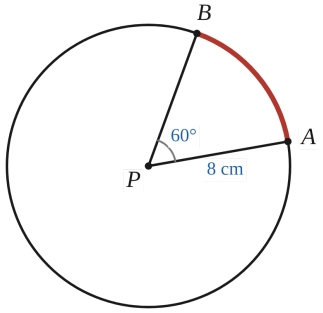
a. Find the circumference of a circle with a diameter of 9 centimeters.

b. A circle has a circumference of 38 feet. Find its radius.

Arc length: arc length of AB = (m(arc AB) ÷ 360°) · _____

Example 2

Find the length of arc AB. Round to the nearest hundredth.



Radians: 180° = _____ radians. To change degrees to radians, multiply by π/180; to change radians to degrees, multiply by 180/π.

Example 3

a. Convert $45°$ to radians.

b. Convert $\frac{5\pi}{6}$ radians to degrees.

Part 2 · Partner practice

Partner practice

a. Convert $\frac{3\pi}{4}$ radians to degrees.

b. Convert $120°$ to radians.